JOBHELPER

JobHelper technical documentation

How the app is built, run and tested

For JobHelper on macOS
10 figures · 25 tables

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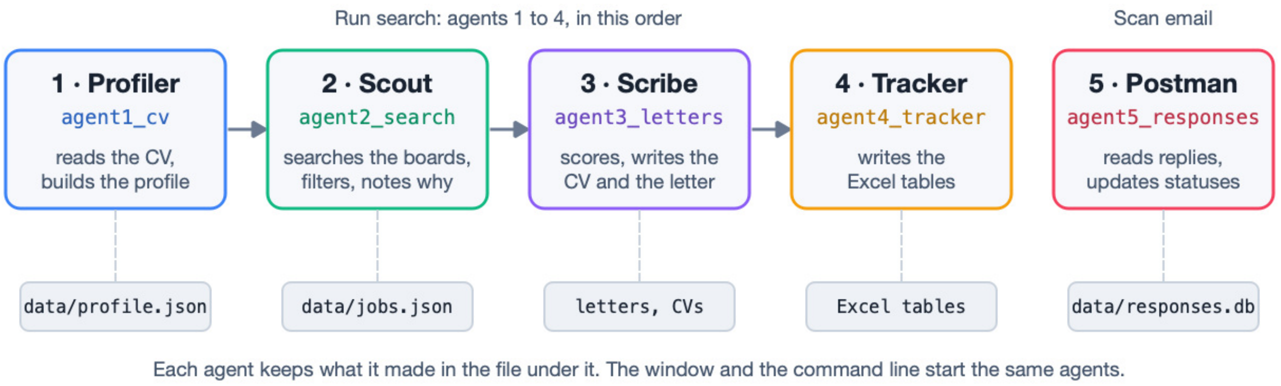
Applies to: the JobHelper source code, on macOS with Python 3.10 or newer.

This document is for people who work with the code. It tells you how to install the app and how to run it from the command line. It also shows how the code is organised, and how to test and document a change. To use the app, read the user guide, GUIDE.md.

1. Overview

JobHelper is a desktop app for macOS. It finds job postings and scores them against a CV. For the good postings, it writes an adapted CV and a cover letter. It also tracks applications and replies. It is written in Python, and the window uses Qt Quick (PySide6). All model calls go through the Claude Code program. So they use the Claude subscription of the person, never the paid API.

The work is done by five agents (Figure 1.1). A search runs the first four in order; an email scan runs the fifth.


Run search: agents 1 to 4, in this order
Scan email
1 · Profiler
agent1_cv
reads the CV, builds the profile
2 · Scout
agent2_search
searches the boards, filters, notes why
3 · Scribe
agent3_letters
scores, writes the CV and the letter
4 · Tracker
agent4_tracker
writes the Excel tables
5 · Postman
agent5_responses
reads replies, updates statuses
data/profile.json
data/jobs.json
letters, CVs
Excel tables
data/responses.db

Each agent keeps what it made in the file under it. The window and the command line start the same agents.

Figure 1.1 – The five agents and what each keeps

Six rules shape the whole code base, and every change must keep them (Table 1.1).

Related documents

Table 1.2 lists the other documents of the project.

Table 1.1 – The rules every change must keep

Rule	What it means in the code
The subscription, never the paid API	llm.ask refuses to run while ANTHROPIC_API_KEY or another setting in config.PAID_SETTINGS is set. The Claude program reads no settings file of the person. No code path calls the API directly
The mailbox is read-only	Folders are opened read-only and messages are fetched with PEEK. Nothing sends, deletes, moves or marks mail
Nothing is claimed that the CV does not show	Every written text is checked in plain code against the CV after the model wrote it
The person's own letters are never changed	Only letters the app wrote may be replaced or moved
Nothing is deleted	Data is backed up before a bulk change; documents are moved to an encrypted backup, old copies to the Trash
No personal data in the code	Names and facts of a real person never appear in code, tests or documents

Table 1.2 – The documents of the project

Document	Use it for
GUIDE.md	Using the app, for someone who never opens a terminal
COMMANDS.md	The full reference of every command and option
HANDBOOK.md	The rules the app is built by, decisions already made, what is left to do
CHANGELOG.md	What changed, day by day. The app shows it under **Help**

2. Know the tech stack

Overview

JobHelper uses a small number of well-known parts. The app must run on the Mac of one person, keep the data private, and use a Claude subscription instead of a paid API. Each part was chosen for one of these needs. This chapter names each part and tells why it was chosen. Then a change does not replace a part without knowing its purpose.

The language and the window

Table 2.1 lists what the app itself is written in.

Table 2.1 – The language and the window

Technology	Used for	Why it was chosen
Python 3.10 or newer	All the logic	The libraries for Word files, Excel files, mail and web pages are mature in Python, and the official Claude library is a Python library. One language for everything keeps the project small enough for one person
Qt Quick (QML) with PySide6	The window	It draws a native window on the graphics card, so the robots and the lists move smoothly. It needs no web browser inside the app. An earlier window was a web page, and the app removed it. Two front ends had to stay the same, and the web window was slower and looked foreign on macOS
QML for layout, Python for logic	The split between ui_qt/ and jobhelper/	A QML file only shows things and calls the bridge; it never reads a file. The logic can therefore be tested without a window, and the window can be tested without real data
Threads with Qt signals	Searches, scans and rewrites	Long work must not freeze the window. A worker thread does it and answers through a signal, which Qt delivers on the window's own thread

Claude

Table 2.2 lists how the app reaches the model.

Documents and data

Table 2.3 lists what reads and writes the person's files.

Table 2.2 – How the app reaches Claude

Technology	Used for	Why it was chosen
claude-agent-sdk driving the Claude Code program	Every question to the model	It uses the person's Claude subscription. The paid API bills per use and would make every search cost money; that is why the app refuses to run while ANTHROPIC_API_KEY is set
One module, llm, for all calls	Asking, counting usage, noticing the limit	With a single door, the rules hold everywhere: no call while the limit is used up, every call counted, no key in an error message. Every call also has a time limit (CALL_TIMEOUT). An empty answer and a signed-out Claude are errors (EmptyAnswer, SignedOut); a signed-out Claude stops a run (STOPS_A_RUN). Text from a web page (a posting, a form's fields) goes in only inside llm.untrusted tags. The system prompt then says llm.POSTING_RULE. A mail has its own tag and rule
Plain code before and after the model	Filters before scoring; checks after writing	A check in code is free, instant and always gives the same answer. The model is asked only what code cannot decide, and what it wrote is checked against the CV in code

Reading the web and the mailbox

Table 2.4 lists what reaches outside the Mac.

Table 2.3 – Documents and data

Technology	Used for	Why it was chosen
python-docx	Reading a CV; writing the adapted CV and the letter	Employers expect Word files. Editing the person's own .docx keeps its fonts and layout, where generating a new file would lose them
pypdf	Reading a CV that is a PDF; joining the letter and the CV into one PDF	It is pure Python, so nothing else has to be installed
LibreOffice (optional)	PDF copies of Word files	It converts a Word file faithfully and is free. It is optional: without it the app still works and offers Word files only
openpyxl	The Excel tables	People already track applications in Excel; the tables can be opened, sorted and shared without the app
JSON files	The job list, the profile, the Options	The data of one person is small. A file can be read whole, backed up by copying, and repaired by hand if it must be
SQLite	The replies read from the mailbox	Replies are many, are looked up by message, and must never be read twice. A small database does that; it is part of Python and needs no server

Privacy and the Mac

Table 2.5 lists what protects the data and ties the app to macOS.

Table 2.4 – Reading the web and the mailbox

Technology	Used for	Why it was chosen
requests with beautifulsoup 4 and lxml	Reading the job boards and company career sites	The boards have no common interface, so their public pages are read as a browser would read them, at a modest pace. No search needs a key or an account
imaplib, from Python itself	Reading replies	IMAP works with almost all mail providers. It can open a mailbox read-only. This keeps the promise of the app: it never sends, deletes or marks mail. email_client.Mailbox is the only way the app reads a mailbox. It opens each folder read-only, reads with PEEK and always signs out. It has no method that writes. Every connection checks the server's certificate (email_client.tls_context), so no other server can get the password
An app password instead of a sign-in window	Access to the mailbox	The person can cancel it at the provider at any time, without a change to the real password. The app needs no registration with each provider

Table 2.5 – Privacy and the Mac

Technology	Used for	Why it was chosen
AES-256-GCM, from cryptography	Encrypting the data files and the backups	Applications and replies are private. This method also notices when a file was changed, so a damaged or tampered file is refused instead of read
The macOS Keychain, through the security program	Keeping the data key and the mailbox passwords	Secrets are then protected by the Mac's login and are never written to a file of the project

Continuation of Table 2.5

Technology	Used for	Why it was chosen
Signed rule files	The rules for CVs and letters, the letter template, the master CV	These files decide what is written in the person's name. A change made outside the app must be approved before the next run
A LaunchAgent	The weekly run	It is how macOS starts a task on a schedule, also when the window is closed
Apple events, through the osascript program	The assistant reads and fills in the form that is open in the browser of the person. This works in Chrome, Brave, Edge, Opera and Safari. Opera GX closed this access, and Firefox never had it	The form is filled in where the person is already signed in, with nothing to install. Chrome and macOS each ask for consent first. The app can do only what a script in the page can do: it cannot read saved passwords. It attaches a file the way a page script can (see section 6)
Hash-pinned packages in requirements.lock	Installing	Each package is checked against a fingerprint, so a changed package cannot be installed unnoticed

Quality

Table 2.6 lists what keeps the code correct.

What is deliberately not used

Table 2.7 lists parts that were considered and left out.

Table 2.6 – Tools for quality

Technology	Used for	Why it was chosen
pytest	More than 2,000 tests	Every change comes with tests. The suite needs no network and never touches real data, so it can be run at any time
pytest-xdist	Running the tests on every core	The suite takes about one minute instead of six. The tests of one window file share one window and follow on from each other. So each window file stays on one worker in its order (xdist_group). The four files run side by side
The window loaded without a screen	Tests of the window	The real QML is loaded and driven, so a broken binding or a missing name fails a test instead of reaching the person
pyflakes	Unused and undefined names	It finds a whole kind of mistake in a second, before a test has to
Git and GitHub	The history of changes, and the public copy of the project	Every finished step is a commit that can be gone back to. Each change is a branch and a pull request. The main branch takes a change only after the tests pass on GitHub Actions. scripts/push_check.py runs before every push and stops one that would publish a personal detail, a secret or the data folder

Next steps

- Install the app

Table 2.7 – What is deliberately not used

Not used	Why not
The paid Claude API	It bills per use. The subscription the person already has does the same work
A web browser inside the app	It was tried and removed: slower, larger, and a second front end to keep the same
A browser extension for filling in forms	It would have to be installed by hand and kept in step with the app. Apple events reach the same page without one
A database server	One person's data fits in files; a server would be one more thing to install and to keep running
A cloud service for the data	Applications, replies and the CV stay on the person's Mac
Sending mail	The app reads replies and drafts texts. The person sends them, so nothing leaves in their name without them

3. Install the app

Prerequisites

Before you install, make sure you have:

- macOS.
- Python 3.10 or newer.
- A Claude subscription. The SDK carries its own copy of the Claude Code program. Sign in one time: select **Sign in** in the Account card, or run claude auth login.
- Optional: LibreOffice, for PDF copies of the documents.

Important
Do not set ANTHROPIC_API_KEY. Leaving it unset is what makes the app use the subscription, and the app refuses to run a search while it is set.

Install the packages

1. Open a terminal in the project folder.
2. Create a virtual environment and turn it on:

```
python3 -m venv .venv
source .venv/bin/activate
```

3. Install the exact, hash-checked packages:

   ```
   pip install --require-hashes -r requirements.lock
   ```

4. Optional: copy the example settings file into the data home. The mailbox password never goes into it.

   ```
   mkdir -p ~/Library/Application\ Support/JobHelper
   cp .env.example ~/Library/Application\ Support/JobHelper/.env
   ```

Table 3.1 lists the main packages and what each is for.

Table 3.1 – The main packages

Package	**Used for**
claude-agent-sdk	Driving the Claude Code program, which uses the subscription
PySide6	The window (Qt Quick and QML) and the PDF of the documentation
python-docx, pypdf	Reading a CV; writing CVs and letters as Word files
openpyxl	The Excel tables
requests, beautifulsoup4, lxml	Reading the job boards
cryptography	AES-256-GCM encryption of the data files and the backups
pytest, pytest-xdist	The tests, on every core. pyflakes, installed separately with pip install pyflakes, checks for unused or undefined names

4. Run the app

Start the window

1. Turn the virtual environment on.
2. Start the app:

   ```
   python app_qt.py
   ```

3. Follow the welcome card. Sign in to Claude, choose a CV, and connect a mailbox for replies if you want. The user guide describes each step.

Build JobHelper.app

A packaged app opens like any other Mac app, without Python or a terminal. To build it:

1. Turn the virtual environment on.

2. Install the build tool. The app itself does not need it.

   ```
   pip install -r requirements-build.txt
   ```

3. Build the app and check it:

   ```
   python scripts/build_app.py
   ```

The script runs PyInstaller with packaging/JobHelper.spec and puts the app in dist/. Then AppCheck examines the app. It looks for the program, the bundle id, the version, the window files, the icon, the sound and CHANGELOG.md. It starts the Claude program that the app carries, and it verifies the signature. Then it starts the app with --check-window. To check an app that is already built, use python scripts/build_app.py --check.

The window check loads the whole window off-screen for a few seconds and reports each problem that Qt finds. It also loads the commands of the weekly run, and saves a picture of the window as dist/window_check.png. The check runs only on a throwaway data folder (JOBHELPER_HOME) and an in-memory keychain (JOBHELPER_VAULT_BACKEND=memory). Without both, the app refuses it. Its bridge is quiet: it asks Claude nothing and gives no reminder.

Some important points:

- The name, version and bundle id are in jobhelper/app_info.py.
- The spec drops the Qt parts that the window does not use, for example the web engine and 3D (UNUSED_QT). Thus the app is about 390 MB, not 650 MB. A test makes sure that no Qt part the window imports is dropped.
- A module that the app loads by its name must be in hiddenimports of the spec. The window check fails if one is missing.
- The app is signed only for the Mac that built it (an ad-hoc signature), and Apple has not notarized it. On another Mac, the person must select **Open Anyway** one time (user guide, section 2).
- Info.plist contains NSAppleEventsUsageDescription, so macOS can ask before the app controls Google Chrome or Terminal.
- Finder and launchd start an app in /. The Claude program watches the folder it works in. In /, macOS then asked for the Desktop, the Downloads, the music, removable disks and the data of other apps. So app_qt.main() changes to config.work_dir(), an empty folder in the data folder, before it does anything. llm.options_for and ClaudeCli.auth_status give the Claude program the same folder.
- The questions of the app start none of the person's claude.ai connectors (mcp_servers={}, --strict-mcp-config and ENABLE_CLAUDEAI_MCP_SERVERS=false in llm.options_for). The questions have no tools, and the connectors cost about 1.5 seconds for each question.

Run a search without the window

A search can also be run from the command line. It does the same work as **Run search** and writes to the same files. Both make one agent2_search.SearchSettings from the Options, and both add what the search found with agent2_search.take_in. A source that cannot be read is named in the log and in the list failed. Then take_in keeps the earlier rows of the Filtered out list. When no source answers, the search raises agent2_search.SourcesDown.

```
python run.py
```

Table 4.1 lists the options used most often.

Table 4.1 – The most used options of run.py

Option	What it does
--where CITY	Searches in another city than the one saved in Options
--source NAME	Chooses the job boards: czech, linkedin, remote, adzuna, jooble or all
--include-remote	Adds the remote boards
--threshold N	Writes documents only from this score
--per-query N	Asks each board for this many postings per role
--all-languages	Keeps postings written in other languages than English
--max-age-days N	Drops postings older than this
--reuse-profile	Does not read the CV again
--skip-search	Scores the postings already in the list, without searching

> **Note**
> Do not run a search from the command line while the window is running one. Both write the same job list.

5. Use the command line

Everything the window does with applications and replies can also be done with status.py. Table 5.1 lists its commands.

Protect and restore the data

The data files are encrypted with a key kept in the macOS Keychain. Table 5.2 lists the actions of python status.py security.

Table 5.1 – The commands of status.py

Command	What it does
python status.py list	Lists every tracked job with its id and status
python status.py apply ID	Marks a new job as applied
python status.py update ID STATUS	Sets the status of a job
python status.py sync	Reads replies in the mailbox and updates statuses. --days N chooses how far back
python status.py responses	Shows the numbers of replies and rejection reasons
python status.py followups	Lists applications with no reply for a number of days
python status.py rebuild	Writes the Excel tables again
python status.py retailor ID	Writes a letter and an adapted CV for one job
python status.py pdf	Makes PDF copies of the documents (needs LibreOffice)
python status.py letters	Takes in letters the person wrote themselves
python status.py prune	Removes new jobs you have already applied to
python status.py repair-emails	Reads email text again, without Claude, and fixes replies that were misread because the email was HTML only
python status.py repair-statuses	Fixes jobs whose status the app does not know
python status.py usage	Shows the Claude usage of the app, in total and per kind of work
python status.py security ACTION	Encryption, Keychain and backup tools (Table 5.2)

Warning
Anyone who has the recovery key can read the data. Never print it into a log, a chat or a file that is kept.

Table 5.2 – The actions of status.py security

Action	What it does
status	Says whether encryption is on and whether a watched file changed
recovery-key	Prints the recovery key. The window shows the same key in the **Account** card
restore-key	Puts a saved recovery key back into the Keychain
approve	Accepts a change to a rule file, the letter template or the master CV as trusted
backup, backup-all	Makes an encrypted backup of the data, or of everything
restore, test-restore	Restores a backup, or proves that a backup can be restored

6. Understand the structure of the code

The code is in three places (Table 6.1).

The agents

Table 6.2 names the module of each agent and what it may ask Claude.

Table 6.1 – The folders of the project

Folder	**What is in it**
jobhelper/	All the logic: the agents, the rules, reading and writing files
ui_qt/	The window: one QML file per window or part of the main screen
tests/	The tests, and the script that takes the pictures of the user guide
scripts/	Tools for the documentation: the PDF builder, the numbering check, the diagram
docs/	The user guide, this document, their PDFs and the pictures
examples/	Example rule files and a list of company career sites, to copy and change

What a search does with a posting

Figure 6.1 follows one posting through a search. All steps before the scoring are plain code. A posting that does not fit the rules never costs a call to Claude. The same is true for a posting from another profession. The app keeps it with the reason, so the person can bring it back.

Table 6.2 – The five agents

Agent	**Module**	**What it does**	**Asks Claude**
Profiler	agent1_cv	Reads the CV and builds the profile and the search roles	Once per CV
Scout	agent2_search	Searches the boards, filters, keeps every dropped posting with its reason. SearchFilters holds the rules of one search in their order	Never
Scribe	agent3_letters	Scores postings five per call; writes the adapted CV and the letter. Scorer is the one way of scoring, for a search and for scoring again	Yes
Tracker	agent4_tracker	Writes the Excel tables	Never
Postman	agent5_responses	Reads replies and files each to its application. ScanStats counts what one scan found and says it at the end. Each email goes to Claude as data (llm.untrusted), so words in it cannot pass for the app's own instructions	For new replies only

How a text is kept inside what the CV shows

Claude writes the adapted CV, the letter, the form answers and the interview answers. Then code checks them (Figure 6.2). A prompt that says "do not invent" is not a guarantee. So the code reads each draft for technologies that the CV does not show. It also looks for terms of the posting stated as experience. The code sends its findings back to Claude once. If a finding is still there, the app keeps it and shows it to the person. It never cuts it out silently. In the interview preparation, small classes do this work. Question, WorkStory, Project, Tool and Skill each clean one part of the reply. Preparation lists everything said in the voice of the person and checks all of it against the CV in one place.

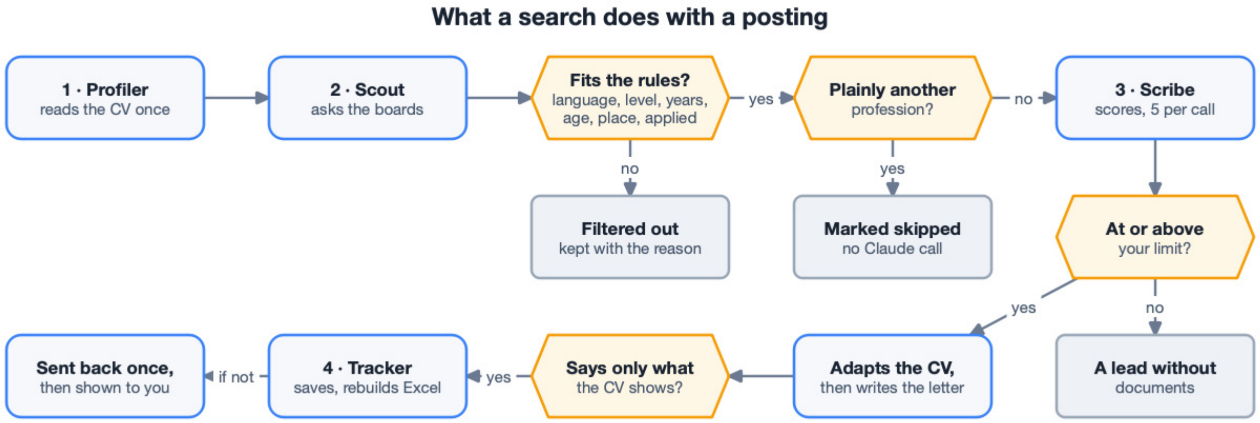

What a search does with a posting
1 · Profiler
reads the CV once
2 · Scout
asks the boards
Fits the rules?
language, level, years, age, place, applied
yes
Plainly another profession?
no
3 · Scribe
scores, 5 per call
no
Filtered out
kept with the reason
yes
Marked skipped
no Claude call
At or above your limit?
yes
no
Sent back once,
then shown to you
if not
4 · Tracker
saves, rebuilds Excel
yes
Says only what
the CV shows?
Adapts the CV,
then writes the letter
A lead without
documents

Figure 6.1 – What a search does with a posting

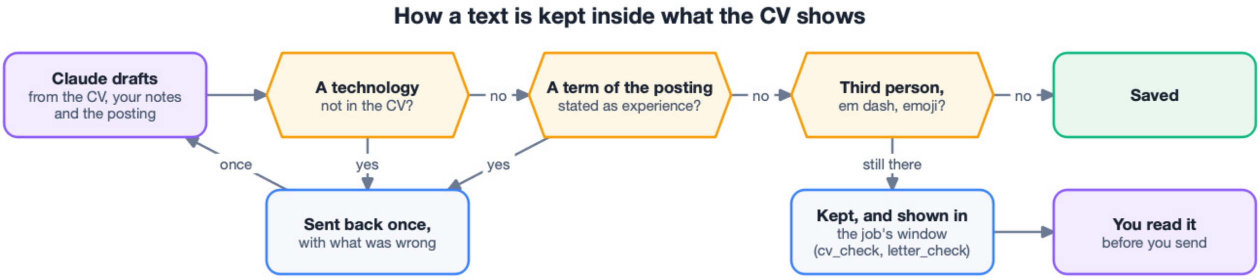

How a text is kept inside what the CV shows
Claude drafts
from the CV, your notes and the posting
A technology
not in the CV?
no
A term of the posting
stated as experience?
no
Third person,
em dash, emoji?
no
Saved
once
yes
yes
still there
Sent back once,
with what was wrong
Kept, and shown in
the job's window
(cv_check, letter_check)
You read it
before you send

Figure 6.2 – How a text is kept inside what the CV shows

What an email scan does with a message

Figure 6.3 shows the way of one message. The mailbox is opened read-only, and rules decide without a call which messages are worth a question to Claude.

How a run behaves at Claude's limit

The limit of the subscription is a state of a run, not an error (Figure 6.4). Every call goes through llm.ask. It refuses to call when a limit is used up. The run then saves its work and shows the reset time. When the person presses **Resume**, the run continues from the same job. A signed-out Claude (llm.SignedOut) also stops the run. The run saves its work and tells the person to log in again. The job keeps its try.

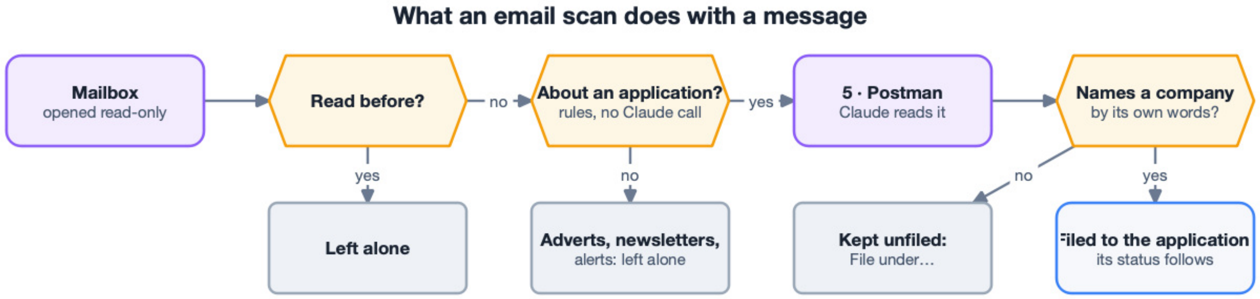

What an email scan does with a message
Mailbox
opened read-only
Read before?
no
About an application?
rules, no Claude call
yes
5 · Postman
Claude reads it
Names a company
by its own words?
yes
no
no
yes
Left alone
Adverts, newsletters,
alerts: left alone
Kept unfiled:
File under…
Filed to the application
its status follows

Figure 6.3 – What an email scan does with a message

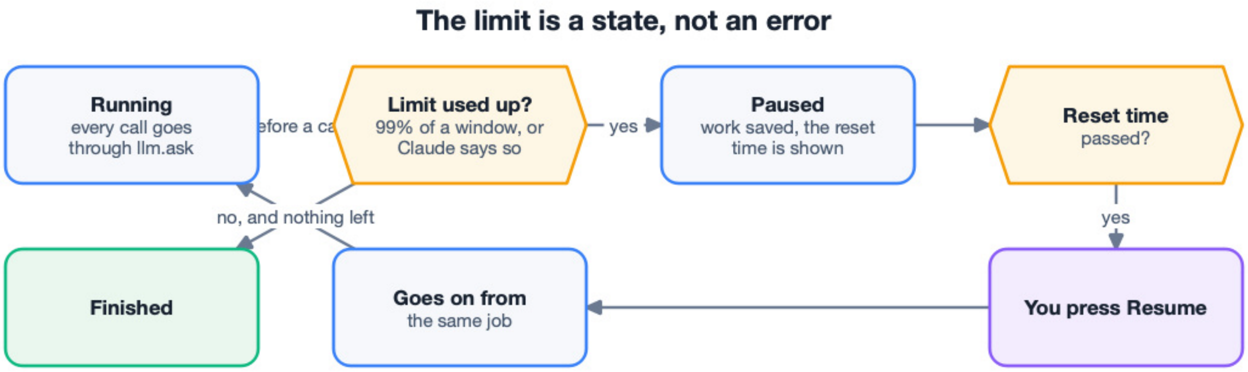

The limit is a state, not an error
Running
every call goes through llm.ask
efore a ca
Limit used up?
99% of a window, or Claude says so
yes
Paused
work saved, the reset time is shown
Reset time passed?
no, and nothing left
yes
Finished
Goes on from the same job
You press Resume

Figure 6.4 – The limit is a state, not an error

The modules around the agents

Table 6.3 groups the other modules by what they are for.

Table 6.3 – The modules around the agents

Group	Modules	What they do
Reading postings	job_sources, scrapers_cz, scrapers_intl, scrapers_remote, scrapers_ats, scrapers_page, web	Each board, feed or API is one JobSource with the same interface: find, reset, available, failures. Each source has one kind; SourceChoice alone decides which kinds a setting reads. One scraper module per kind of site; web says how the app introduces itself to all of them. web also reads a link from a posting only on the public internet (public_address, public_get). web limits a name lookup (LOOKUP_TIMEOUT) and a whole page (public_page: MAX_PAGE, FETCH_TOTAL). jobs.cz and prace.cz share one LmcBoard. A site that turns programs away is left alone
Rules without a model call	filters, prefilter, language, places, keywords, salary, merge, followup, planner	Level, years, language, location, profession, pay, duplicates, reminders. places.RemoteReach decides which remote jobs fit, for the search and for Options. filters.Placement, made once per search, says whether a posting is in the right place and why not. Its home is the country of the search. planner.EventCheck checks a date to remember and names the field of a refusal

Continuation of Table 6.3

Group	Modules	What they do
Writing	word_document, about_you, claims, cv_tailor, cover_letter, letters, form_answers, interview_prep, prep_doc, pdf	The adapted CV, the letter, form answers, the interview documents (one for HR, one technical). WordDocument opens a .docx and saves it with only the text changed; every other part stays byte for byte. claims.ClaimCheck checks that a text in your voice claims nothing more than the CV. Its terms include the posting's skills that the CV does not show (keywords.claimable_missing). A golden set of cases (tests/fixtures/letter_evals.json, run by tests/test_letter_evals.py) measures these checks. The letter, the form answers and the interview preparation all use it. cover_letter.LetterNotes gives the notes a finished letter shows in the job window. claims.EmployerCheck keeps work with its employer. It also finds an employer that no source names (invented). cover_letter.make_template makes a letter template from the CV's head (LetterHead, which also says why it can make none). cv_tailor.MasterSkills keeps the shape of KEY SKILLS. claims.unbacked_figures finds a figure that no source gives; the adapted CV's profile, the letter and the form answers use it. about_you.AboutYou holds the facts of the About you window; WorkNotes makes the notes for each job into the one text that the app reads
Mail	email_client, mailhosts	Reading mailboxes read-only; which server a mailbox is read from. A fetch gives Fetched: the messages, and a warning for each folder or mailbox it could not read

Continuation of Table 6.3

Group	Modules	What they do
The assistant	assistant, assistant_brain, assistant_notes, details, chrome, form_fill, form_run, form_choices, answer_memory	The fixed requests and the facts about the app. What Claude makes of a free request, checked before anything happens. The details of the person. The link to the browser. Reading a form, deciding what each field is, and filling it in. Choosing from lists and menus. The wording of the person, remembered
Claude	llm, usage	The one place that asks Claude; counting usage and knowing the limits. usage.UsageLog keeps the saved sums and the sums of this run; the window asks it for totals and by_tag. usage.windows() gives each limit with the time it was read, and stale when that is more than an hour ago. After a reset the figure is not known (reset_passed), because Claude Code uses the same limit. While the weekly window is used up, the 5-hour one also carries its reset time (held_until). usage.refresh_due() says when the window asks Claude again (REFRESH_AFTER, 10 minutes after the last reading or question, note_try)
Storage and safety	store, db, vault, integrity, housekeeping, health, trouble	The job list, the reply database, encryption, signed rule files, backups, the start-up check, the log of errors

Continuation of Table 6.3

Group	Modules	What they do
The window's back end	ui_api, api_core, api_system, api_options, api_jobs, api_documents, api_replies, api_insights, api_interview, api_letters, api_assistant, api_talk, qt_bridge, bridge_runs, bridge_jobs	What the window calls. Api is a facade: it hands each call to one of ten services. Bridge gives Api to QML and runs long work in threads

The weekly run

The **Weekly run** card writes a LaunchAgent with schedule.write_schedule. The LaunchAgent starts the program itself with --weekly (schedule.command()). From the source, that is python app_qt.py --weekly. In the packaged app, it is JobHelper --weekly. The LaunchAgent also gets a PATH with the usual folders of the Claude CLI and Homebrew.

app_qt.main() sees --weekly before it opens a window and calls weekly.main(). WeeklyRun runs two steps: run.py --reuse-profile, then status.py sync --summary. Each step writes into weekly_jobhelper.log in the data folder. A step that fails is written in the log, and the next step still runs.

The weekly run and the window never change the job list at the same time. store.RunLock is a file lock (.jobs.lock in the data folder). WeeklyRun waits up to WeeklyRun.WAIT seconds for it. If the lock does not come free, the run writes a line in the log and stops with WeeklyRun.BUSY_CODE. The window takes the lock before a search, a scan or a new score, and lets it go when the run ends. store.save_jobs holds the lock while it writes (RunLock.holding()). A change in the window holds it from its first read to its last write (api_base._holding_jobs). While another process holds the lock, both refuse with store.RunLock.BUSY. The commands run.py and status.py hold the lock for their whole run (RunLock.command). Commands that change no job (status._changes_jobs) do not take it.

At each start, schedule.refresh() compares the LaunchAgent with schedule.command(). If an older version or a different place of the app wrote it, the function writes it again with the same day and time. When JOBHELPER_HOME chooses another data folder, the function does nothing and the window shows the weekly run as Off.

The Claude program

The SDK asks Claude through its own copy of the Claude Code program (claude_agent_sdk/_bundled/claude). claude_cli.ClaudeCli finds the same copy. The app uses it to check the sign-in (api_system._claude_auth_status) and to open Terminal on auth login (Bridge.claudeLogin), with the full path of the program. Thus the check, the sign-in and the questions always agree. An app that Finder opens gets a short PATH. If the copy is missing, ClaudeCli looks on the PATH and then in the usual folders. If it finds no program, the window shows claude_cli.MISSING.

Programs the app needs from outside

Table 6.4 lists the programs from outside that the app uses, and what it does without them.

Table 6.4 – Programs from outside the app

Program	Where the app finds it	Without it
LibreOffice	The PATH, then its usual folders (pdf._CANDIDATES), also for an app that Finder opens	No PDF copies. The **PDF copies** row of the Account card says how to get them
The notifications	Contents/Frameworks/libJobHelperNotify.dylib in the built app (macos/Notify.swift, Apple's notification framework, loaded by notify.AppNotifier). macOS asks once and lists JobHelper in System Settings	osascript shows the notification (notify.ScriptNotifier), for python app_qt.py

The check before a push

scripts/push_check.py examines what a git push would publish. It examines only the commits that no remote branch has yet. Before the first push, this is the whole history. A commit on GitHub is public already, also the commit that GitHub makes when it squashes a pull request. As git's pre-push hook, it examines the commits of the branches that the push sends. To install the hook, use python scripts/push_check.py --install-hook. The hook stops the push when it finds one of these problems:

- A personal detail of the owner: the name, an email address, a phone number or a profile address. The check reads them from the master CV and the mailbox settings in the data home. It also reads push_check_words.txt there, with one word on each line. Thus the list is never in the repository. The owner can be named in two places only: the README's "Author" section and the copyright line of LICENSE.md.
- A secret: an API key, a token, a password or a private key. A value with fewer

than 10 different characters is a made-up test value and does not count.

- An author or committer email that is not a GitHub noreply address.
- An assistant credit line in a commit message.
- The data folder or a .env file.

The check also reads the text inside Word files and PDFs. The report gives the kind of problem and its place, but never the private word itself.

The macOS widget

The widget is a WidgetKit extension in Swift: widget/JobHelperWidget.swift. scripts/build_app.py compiles it with the Swift of the Command Line Tools (WidgetBuild). It puts the result into the app as Contents/PlugIns/JobHelperWidget.appex and signs it with widget/JobHelperWidget.entitlements. Then it signs the app again, because the contents of the app changed. Xcode is not necessary.

macOS runs a widget in its sandbox, so the widget cannot open the encrypted data. The entitlements give it read-only access to one folder: ~/Library/Application Support/JobHelper/widget. widget_feed.WidgetFeed writes snapshot.json there. The file holds the counts, the next five dates, the last run, the Claude usage left and the weekly run. It holds no letters, CVs, mail or notes, and its mode is 0600. SystemService.widget_update writes it. The window calls it at start, after each reload and every 5 minutes. weekly.main() calls it after the weekly run.

The widget is linked with -e _NSExtensionMain, as Xcode links an extension. Started in its own main, it stopped at once on macOS 26. The **Open** button and a click open jobhelper://open, which the app registers in CFBundleURLTypes.

The widget reads the file again every 15 minutes, and at midnight it changes "Tomorrow" to "Today". A test makes sure that the widget knows each key that the app writes. To draw the pictures of the user guide, compile with -D PICTURES and scripts/widget_pictures.swift.

The start-up check and its fixes

health.run() makes each check of health.CHECKS. A check returns an empty string when all is well. It returns a sentence when it finds a problem. It returns a pair when the window can fix the problem: the sentence and an action. The actions are in health.ACTIONS: repair_statuses, restore_jobs and restore_replies. The report gives the action of the first problem and the label of its button.

The button calls Bridge.fixProblem. That slot refuses during a run. Then it calls SystemService.fix_problem:

- repair_statuses makes a backup and calls store.repair_statuses. The command python status.py repair-statuses uses the same function.
- restore_data finds the newest backup copy that opens with vault.newest_good_copy. It moves a copy of the damaged file into backups/damaged with vault.set_aside(..., move=False). Then it writes the good copy over the file. If that write fails, the damaged file stays in its place.

No message of the check tells the person to type a command. A test reads the strings of health, vault and agent2_search to make sure.

How the assistant fills in a form

The assistant works on the tab in front in the browser of the person (Figure 6.5). It works in small steps that are easy to check. form_fill holds the scripts and the rules. One fill is one form_run.FormFillRun, with one method for each step. FormChoices makes the choices of a form, and FormFiles gives its file fields their files. form_fill.PageAddress says if the assistant may work on a page at all. A step that cannot go on raises Stop with the message for the person, and the steps after it do not run. When the page was already changed, the report still lists what was filled, and it adds why the fill stopped:

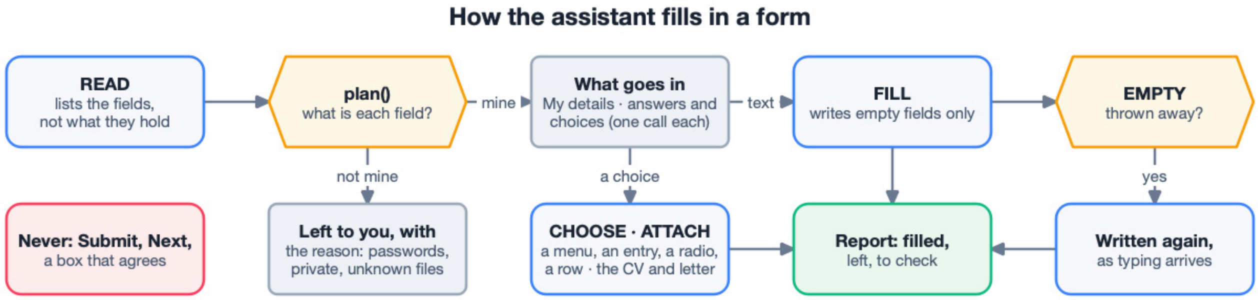

How the assistant fills in a form
READ
lists the fields, not what they hold
plan()
what is each field?
mine
What goes in
My details · answers and choices (one call each)
text
FILL
writes empty fields only
EMPTY
thrown away?
not mine
a choice
yes
Never: Submit, Next, a box that agrees
Left to you, with the reason: passwords, private, unknown files
CHOOSE · ATTACH
a menu, an entry, a radio, a row · the CV and letter
Report: filled, left, to check
Written again, as typing arrives

Figure 6.5 – How the assistant fills in a form

1. A script lists the fields of the page: what each is called, its kind, and whether it already holds something. It numbers them with an attribute and sends back no value the person typed.
2. plan() decides in plain code what each field is. It uses four small classes. Field wraps one field and knows what it asks for. RowTracker knows which job, degree or language a field belongs to. FormPlanner has one method for each kind of field, and FormPlan keeps the result. The details of the person come from details. Open questions go to form_answers in one call. The other fields stay empty, with the reason.
3. A second script writes the values into fields that are still empty and outlines them. Every script gets its values through with_values, as JSON data, so a value never becomes code. Every answer of a page goes through page_reply, because a page may answer anything.
4. Lists, radio buttons and menus that the page draws itself are choices. form_choices decides them. A third script opens a menu, reads its entries, and presses the chosen entry.
5. A file field can ask for the CV, the letter or other attachments (file_kind). It gets the document of the job. A request to the browser has a size limit. So the file goes to the page as text, in pieces of 150,000 characters (PUT). ATTACH makes the pieces into a file. It gives the file to the field the same way as a file chosen by hand. Each piece carries the address of the form. A tab that shows another page gets nothing (PUT answers "moved"). A form left in a tab behind keeps the pieces until it is reloaded.

6. A field that names one of your details gets your value only when it asks about you (_yours). A field about another person (a referee, a spouse) is left to you. So are the salary you get now, a yes/no question and an employer's address. The words of the label come before the names inside the page.
7. Only two kinds of tick box are ticked: "I currently work here" and "I am fluent in this language". The page scripts and the planner read the same words (CURRENT_BOX_WORDS, FLUENT_WORDS, NEVER_TICK_WORDS).

match_job finds the job of a form. A form is usually not at the address of its posting. So the function tries these steps in order:

1. the address of the posting (of the page, or of the page that opened it)
2. the name of the company in the address or the title
3. the company and the job title, both in the words of the page
4. the posting last opened from the app, if the page names its company or its title

If none of these is clear, the function gives no job. Without a job, no file leaves the app.

Table 6.5 gives the rules these steps keep. tests/test_assistant.py checks each of them.

A test area also tries the whole request. tests/fixtures/fill_in_browser.py serves the forms in tests/fixtures/forms/ from this machine. It opens one form in a browser engine and runs the same code as the window. Prepared answers replace Claude, and the data is in a temporary folder. The area waits for the state of the page that it reads, for 10 seconds at most. It does the work one time for each page. When the browser engine stops answering, the area stops in one minute and says why. The area has these forms:

- a plain form
- a form made like component libraries make them. Each box keeps its own value and undoes a change that did not arrive like typing. Its lists are menus that are empty until they open.
- a long form with lists, radio buttons and a number
- a form in a frame from another address

To check a form that the assistant fills in wrong, add a page there. Then add a test that names the fields to fill and the fields to leave.

Table 6.5 – The rules of filling in a form

Rule	How it is kept
The form is never sent	No script presses a button that sends. Only one script presses something (CHOOSE). It accepts five types of element. These are what opens a menu, an entry of a menu, and a radio button. The fourth is a button that adds a row for a job or a degree. The fifth is a tick box. Its whole label must say that a job goes on or that the person is fluent in a language (CURRENT_BOX_WORDS, FLUENT_WORDS). A label with words from NEVER_TICK_WORDS is refused. It reads the words of the element again before it acts. It refuses anything that sends, continues, uploads or agrees. A test reads the scripts for these calls. The test area checks on each form that nothing was sent
What a page throws away is noticed	After writing, EMPTY tells which fields are empty. The app writes these again, the way typing arrives. If a field is still empty, the app tells the person and does not count it as filled
A choice is made only when something settles it	form_choices.settle uses the details of the person and the job. One call to Claude covers the other choices. The app drops an answer that is not one of the options of the field. Every choice is marked for checking
Only the assistant's own answers are read back	To learn the person's corrections, VALUES is asked for the numbers of the fields the assistant filled as answers, and for nothing else
Only empty fields are written	The filling script tests each field again just before it writes
Values are data, never code	They travel as JSON inside the script, and the script travels as an argument of osascript
What the page answers is not trusted	read_answer and fill_answer accept only the expected shape and cut every text to a fixed length

Continuation of Table 6.5

Rule	How it is kept
Private fields are left alone	Passwords, payment and personal-status fields are never planned. The planner finds them by their kind, their label (also in Czech and German) or their autocomplete value (cc-, bday, sex, one-time-code). The review of a form uses the same check. A field of any kind that gives consent (terms, personal data, a background check, a newsletter) is never answered
A file is attached only when it is clearly the right one	The field must ask for a CV, a letter or other attachments. It must be empty and take the type of file. The app must know the job. A photo, a passport or a transcript is never attached. The page must report a file of the same size. If not, the field is listed as left
The right page gets the values	The filling script compares the page's address with the one that was read, and writes nothing when they differ
The browser is never started	The request first asks whether the browser is running and stops there when it is not

How the assistant decides what to do

Figure 6.6 shows what happens to a request.

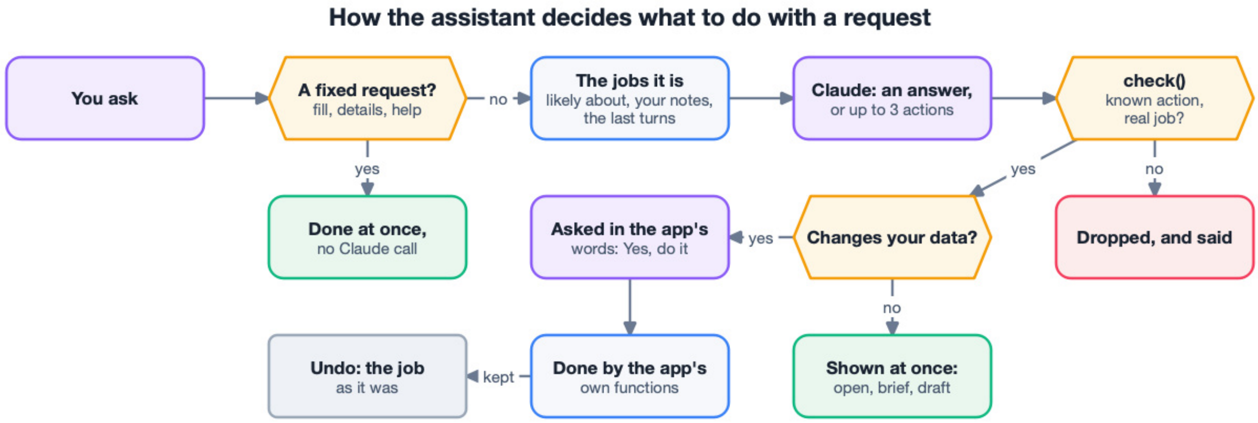

How the assistant decides what to do with a request
You ask
A fixed request? fill, details, help
no
The jobs it is likely about, your notes, the last turns
Claude: an answer, or up to 3 actions
check() known action, real job?
yes
Done at once, no Claude call
yes
no
Asked in the app's words: Yes, do it
yes
Changes your data?
Dropped, and said
no
Undo: the job as it was
kept
Done by the app's own functions
Shown at once: open, brief, draft

Figure 6.6 – How the assistant decides what to do with a request

Some requests are fixed, for example "fill in the form", "my details" or "help". All other requests go to assistant_brain.decide. This makes one call to Claude. The call sends the jobs as lines, the numbers, the last turns of the talk and the list of actions. Claude answers in words or names one action. Table 6.6 gives the rules that keep this safe. api_talk applies them.

Table 6.6 – The rules of acting on a request

Rule	How it is kept
Claude does nothing itself	It returns JSON. check keeps only words, up to three actions from a fixed list, and arguments of the right type. It drops an unknown action or a job that is not in the list
Only the jobs that a request is probably about are sent	relevant picks up to 70 jobs: the job in front, the companies named in the talk, the open jobs and the best leads. Closed applications go only when the request is about them. To send all jobs with each request cost three times more
What was agreed to can be taken back	assistant_confirm keeps the job as it was, and any lead that marking it applied folded away; assistant_undo puts them back. Not offered while a search is running
The prompts were tried against the real Claude	A trial on invented data showed a reply's date being read as an interview's date, and a guessed reason for a score. The job lines were reworded ("an invitation arrived", "no date set") and the rules tightened; run such a trial again after changing a prompt
Nothing of the person's is changed without their yes	An action that changes data comes back to the window as a question, worded by the app from the checked arguments, not by Claude. assistant_confirm carries it out and checks it once more
A rewrite is shown before it is saved	The new paragraph or answer is made first and travels with the question; saving is the confirmed action
An answer comes from what the app holds	The prompt carries the jobs, their state and the facts about the app. A brief, advice, a review and practice feedback each have their own prompt, with only their own facts
The fixed requests cost nothing	They are recognised by their words before any call is made

Continuation of Table 6.6

Rule	How it is kept
Initiative uses no model	The app calculates the briefing and the reminders from the data. Noticing a form is off until the person turns it on. It looks only at the address and title of the tab in front, and at the number of its empty fields

How the window talks to the logic

Figure 6.7 shows the way of one press of a button through the layers and back.

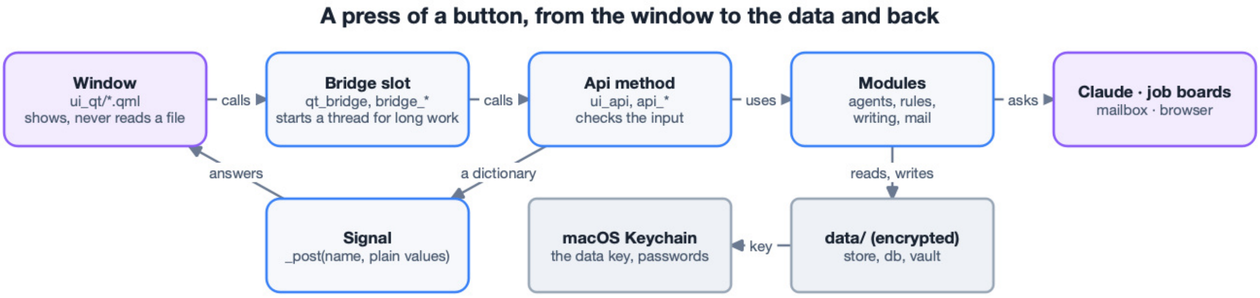

A press of a button, from the window to the data and back
Window
ui_qt/*.qml
shows, never reads a file
calls
Bridge slot
qt_bridge, bridge_*
starts a thread for long work
calls
Api method
ui_api, api_*
checks the input
uses
Modules
agents, rules,
writing, mail
asks
Claude · job boards
mailbox · browser
answers
a dictionary
reads, writes
Signal
_post(name, plain values)
macOS Keychain
the data key, passwords
key
data/ (encrypted)
store, db, vault

Figure 6.7 – A press of a button, from the window to the data and back

The window never touches a file. Every action goes through three layers:

1. A QML file in ui_qt/ calls a slot of bridge, for example bridge.saveOptions(...).
2. The slot, in qt_bridge.py, bridge_runs.py or bridge_jobs.py, calls a method of Api.
3. Api in ui_api.py hands the call to the service that owns it. The service is in one of the api_*.py files. It checks the input and does the work. It returns a plain dictionary: {"ok": True, ...} or {"ok": False, "msg": "why"}.

Long work (a search, a scan, a rewrite) runs in a thread and answers through a signal.

The window keeps no copy of a limit. bridge.limits() gives the length that each typed text may have, and the range of each number box in Options. It also gives the hour and minute of the weekly run. It gives the choices of six lists. They are the kinds of a date, the career-site systems, the Claude models and the text sizes. They are also the cities and the kinds of role that Options suggests. It also gives the limits of a pasted posting and a pasted reply, and the score caps that the hints name. Each number and each list comes from the module that checks it, so the window and the check always agree. The window also asks the back end's own checks through slots. They check an email address, a mail password and another alphabet. They also check a search role, and give the English name of a city.

The back end: a facade over services

Api does no work itself. It follows the object-oriented rules of the project: each class has one job, and parts are joined by composition, not by inheritance. Table 6.7 lists the parts.

Table 6.7 – The parts of the back end

Part	File	What it owns
Api (the facade)	ui_api.py	The list of all calls the window can make. Each call goes to one service
Desktop, CvSource, JobEditor	api_core.py	Opening a file and what to say when it does not open. The CV text. Safe hand edits to the job list, also during a run
SystemService	api_system.py	The sign-in, Claude usage, the log of small problems, backups, the start-up check
OptionsService	api_options.py	Options and the career-site list, and the figures read from them
JobsService	api_jobs.py	The lists, one job in full, hand edits, and the posting last opened
DocumentsService	api_documents.py	Scores, documents on request, documents from a higher score
RepliesService	api_replies.py	Replies from companies and follow-ups
InsightsService	api_insights.py	Insights, the last week, what a search filtered out
InterviewService	api_interview.py	Dates, form answers, interview preparation
LettersService	api_letters.py	The letter editor and PDFs
AssistantService, TalkService	api_assistant.py, api_talk.py	The assistant: forms in the browser, and free requests

Each service gets four things in its constructor: the facade, the JobEditor, the CvSource and the Desktop. A service that needs another service asks through the facade. No service imports another service; a test checks this. A test replaces a class (Desktop.open, CvSource.text) or a call of the facade, never the inside of a service.

Important
Every input is checked twice: in the window, where the person sees a red message next to the field, and again in the Api method, so that nothing unusable is saved whatever sent it.

7. Work with the data files

Figure 7.1 shows how the app keeps the data safe. Every read and write goes through vault. The key is in the Keychain. The app makes a backup before a bulk change. A rule file changed outside the app must be approved.

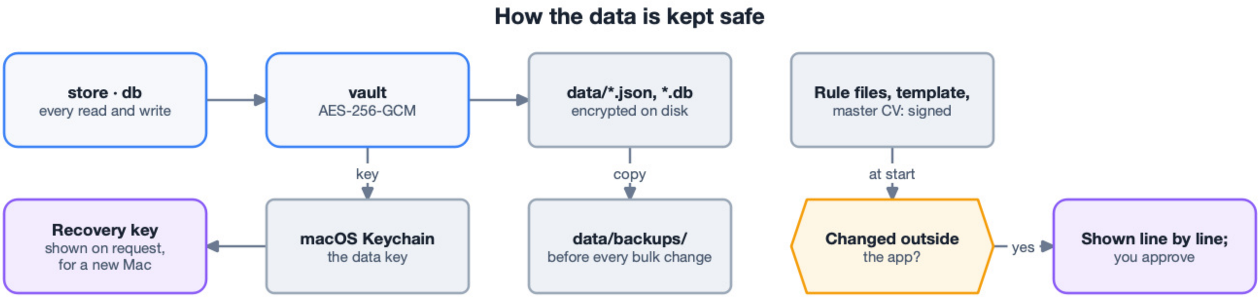

How the data is kept safe
store · db
every read and write
vault
AES-256-GCM
data/*.json, *.db
encrypted on disk
Rule files, template,
master CV: signed
key
copy
at start
Recovery key
shown on request,
for a new Mac
macOS Keychain
the data key
data/backups/
before every bulk change
Changed outside
the app?
yes
Shown line by line;
you approve

Figure 7.1 – How the data is kept safe

Everything a person makes with the app is in the data home (Table 7.1): ~/Library/Application Support/JobHelper. The folder data in it holds the files, and .env the settings that are not secret. home.DataHome gives these places to config. The data home is not part of the code, so a new version of the program does not touch it.

Older versions kept data/ and .env in the project folder. app_qt.main() calls home.start() before it uses the data. If the old folder holds data, DataHome.settle() moves it once:

1. It copies the folder into data.incoming in the data home.
2. It compares each copy with its original, byte for byte. A difference stops the move.
3. It gives the copy its real name and writes the note home.json.
4. It renames the old folder to data_moved_to_Library, and .env to .env_moved_to_Library. Nothing is deleted.
5. The app starts again (home.restart_command), so every module uses the new place.

If a step fails, the old folder stays in use and the start-up check shows the reason. The variable JOBHELPER_HOME names another data home; the tests set it to a temporary folder. With that variable set, the app does not read, change or remove the weekly run of the Mac (schedule.problem). The LaunchAgent always starts the app on the usual data home. In Table 7.1, data/ is the folder data of the data home.

Follow these rules when code changes data:

1. Read and write the job list only through store, and the replies only through db.

Read and write any encrypted file only through vault. Every small JSON file goes through one vault.JsonFile. Examples: the assistant notes, the remembered answers and the notes per company. The Options, the last run, the filtered-out jobs, the CV analysis and My details use it too. read gives the saved value, or nothing when the file is not there or is damaged. load stops with an error for a damaged file. If the key cannot open the file, both stop with an error. Thus a save never writes an empty store over the data. A store that may show nothing in that case catches the error itself. A change starts from load_for_change. It moves a damaged file to backups/damaged first, so the change never writes over it. The clean-up of old backups removes only folders with a time in their name.

2. Call vault.make_backup(label) before a change that touches many jobs.
3. Move a document to the encrypted backup instead of deleting it; put old copies in the Trash.
4. Do not change the data while the window is running.

Table 7.1 – The files and folders in data

Path	What it holds	Encrypted
data/jobs.json	Every posting and application	Yes
data/profile.json	The profile read from the CV	Yes
data/ui_settings.json	The Options	Yes
data/responses.db	Every reply that was read	Yes
data/usage.json	The Claude usage and the last known limits	No
data/backups/	Backups, made before any bulk change	Yes
data/cv/, data/my_letters/	The master CV and the person's own letters	No, so that Word can open them
data/tailored_cvs/, data/cover_letters/	The documents the app wrote	No, so that Word can open them
data/app_trouble.log	One line for each error the app carried on after	No

8. Test a change

Run the tests

1. Run the whole suite. It uses every core of the computer, takes about one minute and needs no network:

```
python -m pytest -q
```

2. Check for unused and undefined names:

```
python -m pyflakes jobhelper tests scripts
```

Both must pass before a change is committed.

What the tests may never touch

The tests work in temporary folders with stand-ins. Table 8.1 lists what they never reach and what stands in for it.

Table 8.1 – What the tests never touch

Never touched	What stands in
The real data/ folder and .env	Temporary folders, set by the isolate_data fixture
The macOS Keychain	An in-memory keychain
Claude	A function that returns a prepared answer
Job boards and mail servers	Functions that return prepared postings and messages
Notifications, Finder, the Trash, the browser	Functions that only record that they were called
The person's browser	A guard that fails the test. The scripts that read and fill in a form run in a browser engine on a local page, tests/fixtures/forms/

Test the window

The tests in tests/test_window_*.py load the window without a screen and drive it. Each of these files gets its own window, so the four files run at the same time. Use the helpers of their ui fixture. ui.js(code) runs JavaScript in the window. ui.find(name) finds an item by its objectName. ui.wait(ms) lets the window run.

> **Tip**
> After a change that can be seen, take a picture of the window and look at it. A test that reads a text does not notice that the text is not drawn.

9. Update the documentation

Each change that a person can see needs three things. Add a line to CHANGELOG.md, and update the user guide. If a window in a picture changed, take its pictures again.

Take the pictures of the user guide again

The pictures show the real window filled with invented people and companies. A script takes them:

```
python -m pytest tests/guide_shots.py -q -o addopts=""
```

The script draws the window at twice its size, as on a Retina screen. Thus the pictures are sharp on large screens and in the PDF.

Keep the numbers of figures and tables in order

Figures and tables are numbered inside their chapter, as in a thesis. For example, the second figure of chapter 4 has the number 4.2. Each picture has a caption under it, and each table has a caption above it. The caption is a line in italics: the word Figure or Table, the number, an en dash and the title. The text refers to each figure and table by its number.

1. Write the caption of a new figure or table with any number that is not used yet, for example 900.
2. Renumber the captions and every reference in one go:

   ```
   python scripts/guide_number.py
   ```

Build the PDFs

Figure 9.1 shows how a PDF is made from its text.

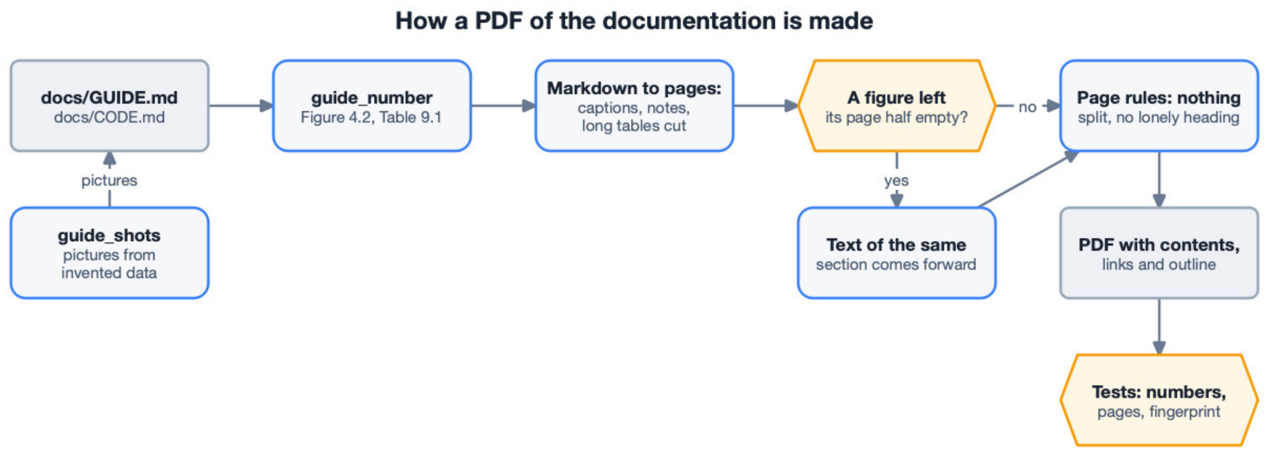

How a PDF of the documentation is made
docs/GUIDE.md
docs/CODE.md
guide_number
Figure 4.2, Table 9.1
Markdown to pages:
captions, notes,
long tables cut
A figure left
its page half empty?
no
Page rules: nothing
split, no lonely heading
pictures
guide_shots
pictures from
invented data
yes
Text of the same
section comes forward
PDF with contents,
links and outline
Tests: numbers,
pages, fingerprint

Figure 9.1 – How a PDF of the documentation is made

```
python scripts/guide_pdf.py
```

The script makes docs/GUIDE.pdf and docs/CODE.pdf. It lays out each one like a thesis. First comes a title page. Then come the contents, with dotted lines to the page numbers. Then comes the text, with the page number at the top right. You can select the entries of the contents, the references to figures and tables, and the links to other chapters. Each one takes you to its place. A caption is centred under its picture and above its table. There are no lists of figures and tables. The script keeps the page rules in Table 9.1. A test fails while a PDF is older than its text.

Write in the house style

Table 9.2 gives the rules for the text of both documents.

Table 9.1 – The page rules of the PDFs

Rule	How the script keeps it
A figure is never split across two pages	A picture and its caption are one block; a block that would cross a page starts on the next one
A table stays on one page	The same, with the caption above it
No page is left half empty because a figure or a table had to start on the next page	The text that follows it in the same section moves forward to fill the space. A figure, a table or a note moves forward only if all of it fits. Text never goes past a heading. A table longer than a page starts where its page has room. A picture that almost fits is made a little smaller
What names a place leads to it	A PDF library adds a link for each entry of the contents and each mention of a figure or a table. It also adds a link for each reference to another chapter. It also writes the chapters into the outline of the PDF
The points of a list stand out	A heavy square before each point, a bold number before each step
A table longer than a page continues between two rows	It is cut into parts that fit, each under the line "Continuation of Table" with its number, and with the heading row repeated
A heading is never left alone at the bottom of a page	A heading whose text would start on the next page moves there too

Table 9.2 – The rules for writing the documentation

Element	Rule
Headings	Start with a verb and name the task: "Connect your mailbox"
Steps	Numbered, one action each, in the order they are done
Reference material	In a table, with a caption and a reference from the text
Names of buttons and labels	Bold, spelled exactly as in the app
Commands and file names	Monospace. The user guide contains no commands
Links	The link text says where it leads, never just "here"
Notes	Four kinds: Note, Tip, Important, Warning, as a quoted block that starts with the bold word

10. Troubleshoot problems

Table 10.1 lists the problems met most often while working on the code.

Table 10.1 – Problems while working on the code

What you see	Cause	What to do
"Unset ANTHROPIC_API_KEY to use your subscription"	The paid key (or the setting the message names) is set in the shell	Run unset with that name, then start again
The window says Claude is not signed in	The Claude Code program has no session	Run claude auth login
A test of a PDF fails after a change to a document	The PDF is older than its text	Run python scripts/guide_pdf.py
A test of the numbering fails	A figure or table was added or moved	Run python scripts/guide_number.py
A QML property named onSomething does nothing	QML reads such a name as a signal handler	Give the property another name
An item made by a Repeater is not found in a test	It is not a child in the object tree	Use ui.seen(part) or ui.drawn(name)
The window stutters while a thread works	Python runs one thread at a time	Give the thread short rests, as keywords.read_ahead does
Data looks wrong after an experiment	A change was made to the real data	Restore the newest backup: python status.py security restore